

Evaluation of sample pooling in SARS-CoV-2 testing using the TaqPath™ COVID-19 Combo Kit

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Summary

The worldwide spread of SARS-CoV-2 has led to a high demand for COVID-19 molecular testing. However, mass testing is causing heavy laboratory backlogs, reagent shortages and laboratory personnel burnouts. To overcome these difficulties, testing samples in pools has been shown to be an effective strategy for low COVID-19 prevalence regions. In the present study, we evaluated the effectiveness of pooled nasopharyngeal specimens testing using the TaqPath COVID-19 Combo Kit. Four confirmed positive samples were tested individually and also used as positive spike-ins for the pooling protocol. Each one was pooled with four different confirmed negatives to a total of five samples per pool. Five microliters of RNA were used for each reaction, whereas reactions with increased RNA input (10 and 15 ul) were performed to investigate probable effects on sensitivity and assay performance. All samples which were tested either individually or in pools resulted positive. Our results showed that testing in

pools of five samples using the TaqPath COVID-19 Combo Kit is a reliable option for SARS-CoV-2 detection, especially when mass testing is urgently needed.



Key words

SARS-CoV-2; COVID-19; pooling; molecular test;
Real-time RT-PCR

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Introduction

Since December 2019, when a novel coronavirus later named Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), was identified as the causative agent of Coronavirus disease 2019 (COVID-19) there have been 56,261,952 confirmed cases including 1,349,506 deaths reported to WHO up to 20 November 2020 (World Health Organization. 2020. Novel Coronavirus (COVID-19), <https://covid19.who.int/>). In order to control the spread of the disease by setting appropriate quarantine measures, rapid diagnosis is needed. Consequently, the high demand for SARS-CoV-2 molecular testing during the pandemic leads to reagent and consumable shortages, while the laboratory personnel is not enough to meet the needs of mass testing. Testing samples in pools was shown to be an effective strategy to overcome these difficulties. If the pool test is negative, all patients whose samples are included in the pool are considered COVID-19 negative, while if the result of the pool is positive, the samples are tested individually. In the present study, we evaluated the effectiveness of molecular testing in pooled nasopharyngeal specimens using the TaqPath COVID-19 Combo Kit (Thermo Fisher Scientific, MA).

Materials and methods

Twenty swabs in viral transport medium submitted for

COVID-19 testing to the Department of Microbiology in Aristotle University of Thessaloniki were included in the study: sixteen were previously determined to be negative using the study's protocol, while four confirmed positive samples were used as positive spike-ins for the pooling protocol. Specifically, each pool contained 400 microliters from each of four negative samples and one positive sample, resulting in a pool with a final volume of 2 ml.

RNA was extracted from 400 microliters of each positive individual sample ($n=4$), and from each pool ($n=4$) using the KingFisher Flex platform (Thermo Fisher Scientific, MA) and a commercially available kit (MagMax Viral/Pathogen Nucleic Acid Isolation Kit; Thermo Fisher Scientific, MA) following the CE/IVD and FDA emergency use authorization protocol provided by the manufacturer.

The RNA samples from the individual positive specimens and the respective pools were subjected to reverse transcription-polymerase chain reaction (RT-PCR) amplification using a commercially available multiplex real-time RT-PCR kit (TaqPath COVID-19 Multiplex diagnostic solution, Thermo Fisher Scientific, MA) following the CE/IVD and FDA emergency use authorization protocol provided by the manufacturer. The kit contains three primer/probe sets specific to different SARS-CoV-2 genomic regions [orf-1ab, spike (S) protein and nucleocapsid (N) protein], as well as primers/probes for bacteriophage MS2 to be used as control RNA to verify the efficacy of the sample preparation and the absence of inhibitors in the PCR reaction. The MS2 phage control

is added to the samples before the extraction of RNA. According to the manufacturers, the limit of detection of the kit is 250 genomic copy equivalents (GCE) per ml of specimen (bronchoalveolar lavage or nasopharyngeal swab) or 10 GCE per reaction, while the *in silico* analysis indicated that significant amplification of non-target sequences resulting in cross-reactivity or potentially interference with detection of SARS-CoV-2 are not likely to occur.

Five microliters of RNA were used for each reaction as per kit recommendation. Furthermore, we performed reactions with increased RNA input (10 and 15 microliters) to investigate probable effects on sensi-

tivity and assay performance. PCR testing was performed on Quantstudio 5 0.2ml platform (Applied Biosystems, Thermo Fisher Scientific, MA). A sample was considered positive when amplification of at least two targets was detected.

Results and discussion

All known positive samples which were tested either individually or in pools using a variety of RNA quantities per sample (5ul, 10ul and 15 ul) resulted positive (Table 1). The cycle threshold (Ct) values of the single

Table 1 Cycle threshold (Ct) values of known SARS-CoV-2 positive samples tested individually and in pools and deviations per target gene

	ORF1ab			N		S		MS2	
	Way of test (RNA in ul)	Ct	individual-pool Ct difference	Ct	individual-pool Ct difference	Ct	individual-pool Ct difference	Ct	individual-pool Ct difference
Sample A	Single (5)	20.47		21.33		20.91		26.76	
	Pooled (5)	24.02	3.55	24.76	3.43	25.56	4.65	27.02	0.26
	Pooled (10)	23.02	2.55	23.79	2.46	23.53	2.62	26.4	-0.36
	Pooled (15)	19.70	-0.77	22.49	1.16	21.47	0.56	22.54	-4.22
	AVEDEV	1.72	1.70	1.18	0.79	1.68	1.37	1.57	1.85
Sample B	Single (5)	17.72		18.41		18.29		34.98	
	Pooled (5)	18.85	1.13	19.57	1.16	19.4	1.11	27.16	-7.82
	Pooled (10)	17.93	0.21	18.48	0.07	18.31	0.02	26.61	-8.37
	Pooled (15)	17.39	-0.33	18.06	-0.35	17.79	-0.5	24.69	-10.29
	AVEDEV	0.44	0.53	0.47	0.58	0.48	0.60	3.31	0.98
Sample C	Single (5)	30.68		31.42		30.87		28.26	
	Pooled (5)	34.05	3.37	34.71	3.29	34.22	3.35	27.5	-0.76
	Pooled (10)	31.97	1.29	32.94	1.52	32.5	1.63	26.77	-1.49
	Pooled (15)	32.08	1.40	32.19	0.77	32.03	1.16	25.07	-3.19
	AVEDEV	0.93	0.90	1.01	0.95	0.95	0.87	0.98	0.92
Sample D	Single (5)	20.27		20.86		20.87		25.38	
	Pooled (5)	22.09	1.82	22.53	1.67	22.57	1.7	25.29	-0.09
	Pooled (10)	21.09	0.82	21.42	0.56	21.45	0.58	24.09	-1.29
	Pooled (15)	20.49	0.22	20.87	0.01	20.94	0.07	22.58	-2.8
	AVEDEV	0.61	0.58	0.56	0.62	0.56	0.61	1.00	0.94
Total AVEDEV			1.27		1.31			1.41	

positive samples ranged from 17.72 to 31.42. The Ct difference between unpooled and pooled samples ranged from -0.77 to 4.65. The mean deviation value (AVEDEV) in the pooled samples was similar in the three target genes (1.27, 1.31 and 1.41 in the Orf1b, N and S genes, respectively).

In one sample (sample B) the Ct values in all target genes of the pool in which 15ul RNA was applied were lower than those in the unpooled sample; a similar finding was observed in sample A, but only in one target gene (Orf1b). It has been hypothesised that the lower Ct values of pools than of single samples occur because of the carrier effect of the higher RNA content in pools.¹

Since many years, testing of pooled samples for various pathogens has been successfully applied. Nowadays, the strategy became more effective due to the development of highly sensitive molecular assays. As the demand for more SARS-CoV-2 testing is increasing, there are reports of efficient pool sample testing.¹⁻⁶ The pool testing procedure is beneficial when the prevalence of the disease in a region is low, because otherwise the number of positive pools which should be tested separately would be high.

A major concern of pool testing is to avoid the false negative tests. A predictive algorithm indicated a pooling ratio of 1 to 5 was expected to retain accuracy.² However, sufficient diagnostic accuracy has been reported even when up to 30 samples of asymptomatic people are included in the pool.¹ It was shown that specimen pooling did not affect the sensitivity of detecting SARS-CoV-2 when the PCR Ct of the original specimen is lower than 35.⁷ Laboratories have to validate the procedure prior to application, to ensure the adequate performance of both the extraction and amplification assays without losing accuracy.

The results of the current study indicate that testing in pools of 5 samples using the TaqPath™ COVID-19 Combo Kit is a reliable option for SARS-CoV-2 detection. This method can be used when mass testing is urgently needed in order to reduce the heavy laboratory backlog, limiting the testing time and the overall cost.

Conflict of interest

The authors declare no conflict of interest.

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Selidis S.A. supplied the extraction and PCR kits for this study.



Περίληψη

Αξιολόγηση της μεθόδου δεξαμενοποίησης δειγμάτων SARS-CoV-2 με τη χρήση του Taq-Path™ COVID-19 Combo Kit

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Η παγκόσμια εξάπλωση του SARS-CoV-2 οδήγησε σε υψηλή ζήτηση για μοριακές τεχνικές ανίχνευσης του ιού. Παράλληλα, ο έλεγχος μεγάλου αριθμού δειγμάτων έχει σαν αποτέλεσμα μεγάλο όγκο δουλειάς για τα εργαστήρια και καθυστερήσεις, ελλείψεις αντιδραστηρίων και εξάντληση του προσωπικού. Για να ξεπεραστούν αυτές οι δυσκολίες, η εξέταση πολλών δειγμάτων μαζί (ομαδοποίηση, pooling) αποδείχθηκε αποτελεσματική στρατηγική για περιοχές με χαμηλό επιπολασμό της COVID-19. Στην παρούσα μελέτη, αξιολογήθηκε η αποτελεσματικότητα ελέγχου ρινοφαρυγγικών δειγμάτων μετά από ομαδοποίηση χρησιμοποιώντας το TaqPath COVID-19 Combo Kit. Τέσσερα επιβεβαιωμένα θετικά δείγματα δοκιμάστηκαν τόσο μεμονωμένα, όσο και μαζί με άλλα στο πρωτόκολλο ομαδοποίησης. Κάθε ένα από αυτά ελέγχθηκε σε συνδυασμό με τέσσερα διαφορετικά επιβεβαιωμένα αρνητικά σε ένα σύνολο πέντε δειγμάτων ανά ομάδα. Πέντε μικρολίτρα RNA χρησιμοποιήθηκαν για κάθε αντίδραση, ενώ πραγματοποιήθηκαν και αντιδράσεις με αυξημένη ποσότητα RNA (10 και 15 ul) για τη διερεύνηση πιθανών επιδράσεων στην ευαισθησία και την απόδοση της ανάλυσης. Όλα τα δείγματα που δοκιμάστηκαν είτε μεμονωμένα είτε σε ομάδες έδωσαν θετικό αποτέλεσμα. Τα αποτελέσματά μας έδειξαν ότι ο έλεγχος σε ομάδες πέντε δειγμάτων χρησιμοποιώντας το TaqPath COVID-19 Combo Kit είναι μια αξιόπιστη επιλογή για την ανίχνευση SARS-CoV-2, ειδικά όταν απαιτείται επείγοντως μαζικός έλεγχος πολλών δειγμάτων.



Λέξεις κλειδιά

SARS-CoV-2, COVID-19, ομαδοποίηση, πραγματικού χρόνου RT-PCR



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